



*WRITE CODE FROM SCRATCH IN A
CLEAR & CONCISE WAY, WITH A
COMPLETE BASIC COURSE.*

*FROM BEGINNERS TO
INTERMEDIATE.*

Python for Beginners with applications

Lesson 4 :
Visualizing Data (matplotlib,
Seaborn, Plotly)
Numpy Library

Abdesselam Filali 2025-03-12 17h45 – 18h30 Algiers time

NumPy

NumPy is a fundamental library for numerical computing in Python. It provides support for large, multi-dimensional arrays and matrices, along with a wide range of mathematical functions to operate on these arrays efficiently.

NumPy

```
import numpy as np

# Creating a NumPy array
arr = np.array([1, 2, 3, 4, 5])

# Performing mathematical operations on the array
result = arr * 2
print(result) # Output: [2 4 6 8 10]
```


PYTHON LIBRARIES FOR DATA VISUALIZATION

- **Matplotlib:** The foundational plotting library, used for basic charts.
مكتبة الرسم البياني الأساسية، وتستخدم للمخططات الأساسية.
- **Seaborn:** Built on top of Matplotlib, ideal for statistical plots.
وهي مثالية للبيانات الإحصائية matplotlib مبنية فوق
- **Plotly:** Used for interactive and web-based visualizations.
تُستخدم للبيانات التفاعلية بيانات الويب

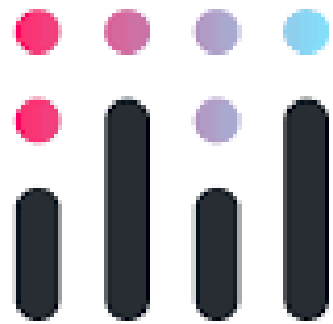
matplotlib

https://matplotlib.org/stable/api/_as_gen/matplotlib.pyplot.plot.html



<https://seaborn.pydata.org/>

seaborn

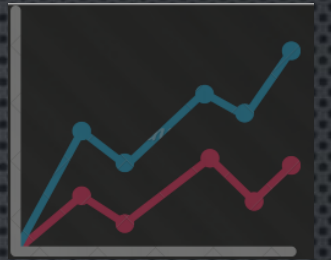


plotly

<https://plotly.com/python/>

Use Case: Showing trends over time.

LINE PLOTS



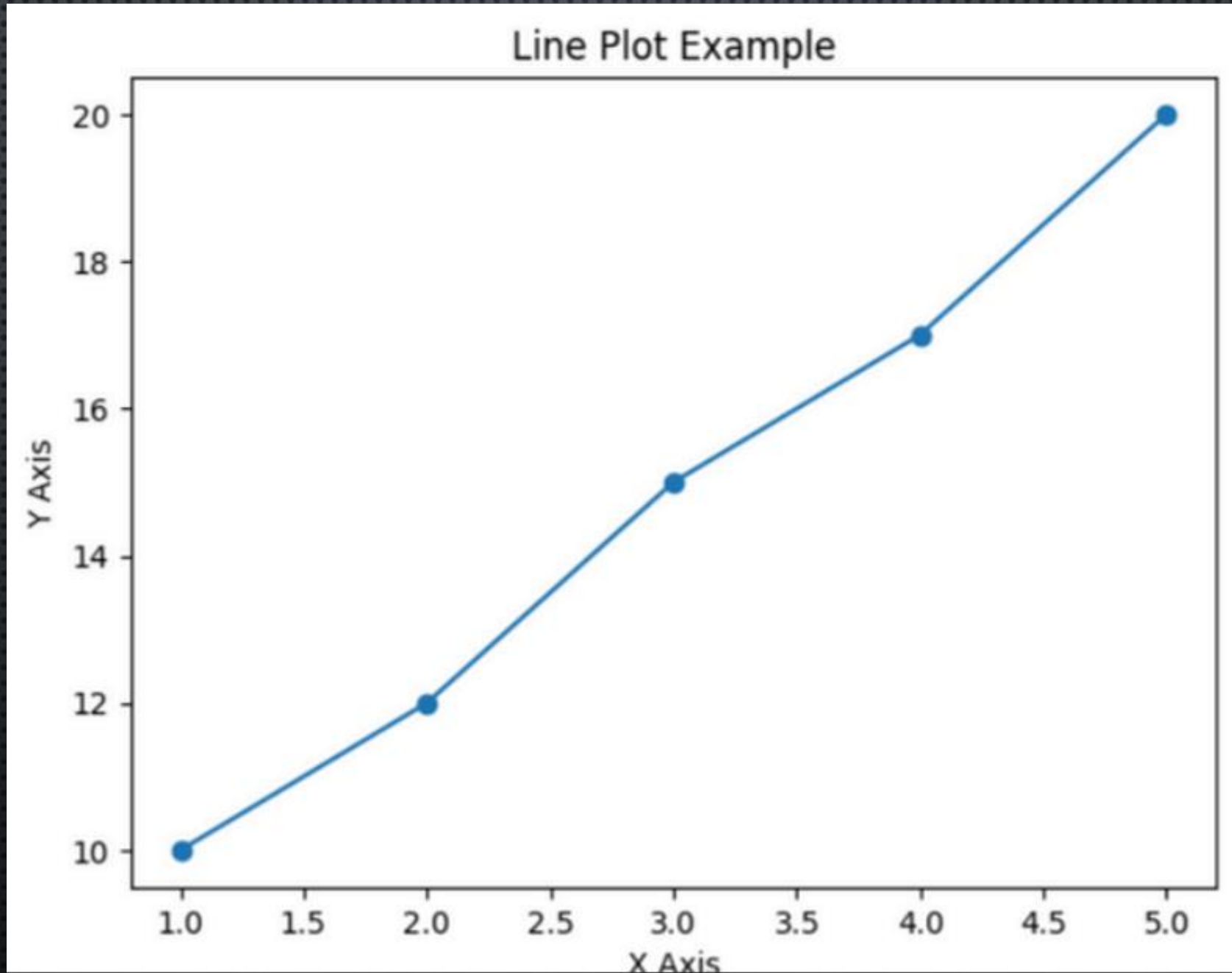
Library: Matplotlib

```
import matplotlib.pyplot as plt # Import the Matplotlib library

# Data
x = [1, 2, 3, 4, 5] # X-axis data
y = [10, 12, 15, 17, 20] # Y-axis data

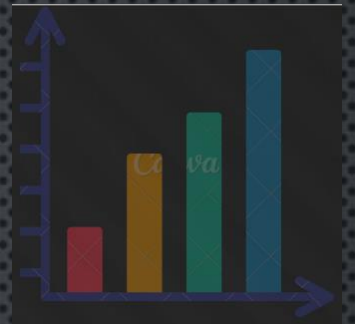
# Create Line Plot
plt.plot(x, y, marker='o') # Plot x and y with circle markers on each point
plt.title('Line Plot Example') # Add a title to the plot
plt.xlabel('X Axis') # Label for the X-axis
plt.ylabel('Y Axis') # Label for the Y-axis
plt.show() # Display the plot
```

LINE PLOTS



Use Case: Comparing different categories or discrete items.

BAR PLOTS

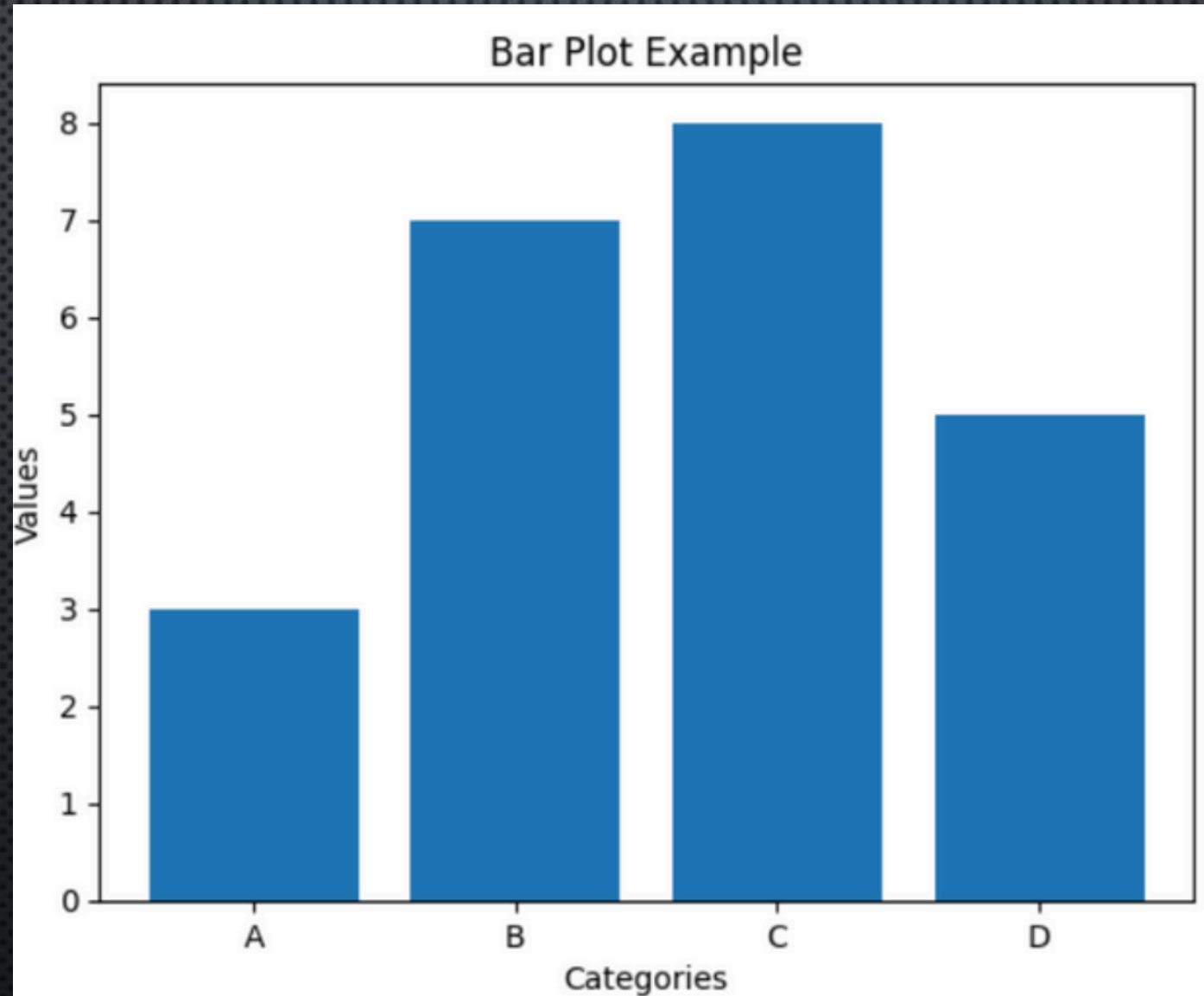


```
import matplotlib.pyplot as plt # Import the Matplotlib library

# Data
categories = ['A', 'B', 'C', 'D'] # Define categories
values = [3, 7, 8, 5] # Corresponding values for each category

# Create Bar Plot
plt.bar(categories, values) # Create a bar plot using categories and values
plt.title('Bar Plot Example') # Add a title to the plot
plt.xlabel('Categories') # Label for the X-axis
plt.ylabel('Values') # Label for the Y-axis
plt.show() # Display the plot
```

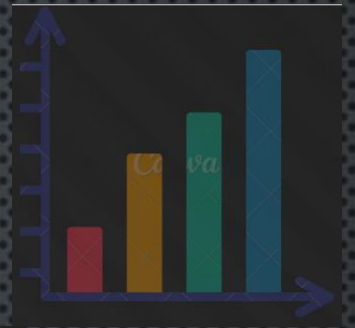

BAR PLOTS



Use Case: Comparing different categories or discrete items.

Library: Matplotlib

BAR PLOTS



```
import numpy as np
import matplotlib.pyplot as plt

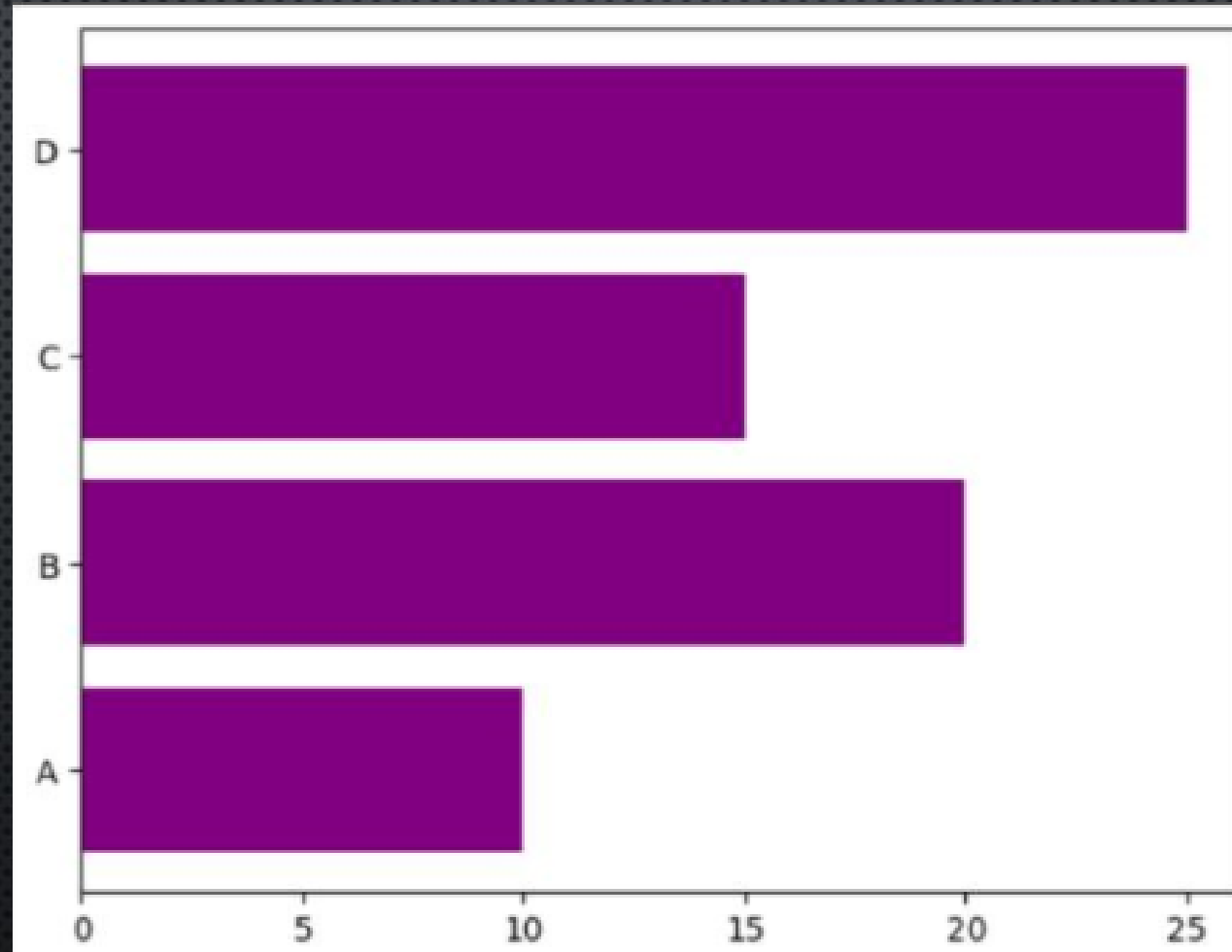
categories = ['A', 'B', 'C', 'D']
values = [10, 20, 15, 25]

plt.barh(categories, values, color='purple')

plt.show()

# Source Code --> clcoding.com
```

BAR PLOTS



Use Case: Showing relation between 2 variables. **SCATTER PLOTS**

Library: Matplotlib

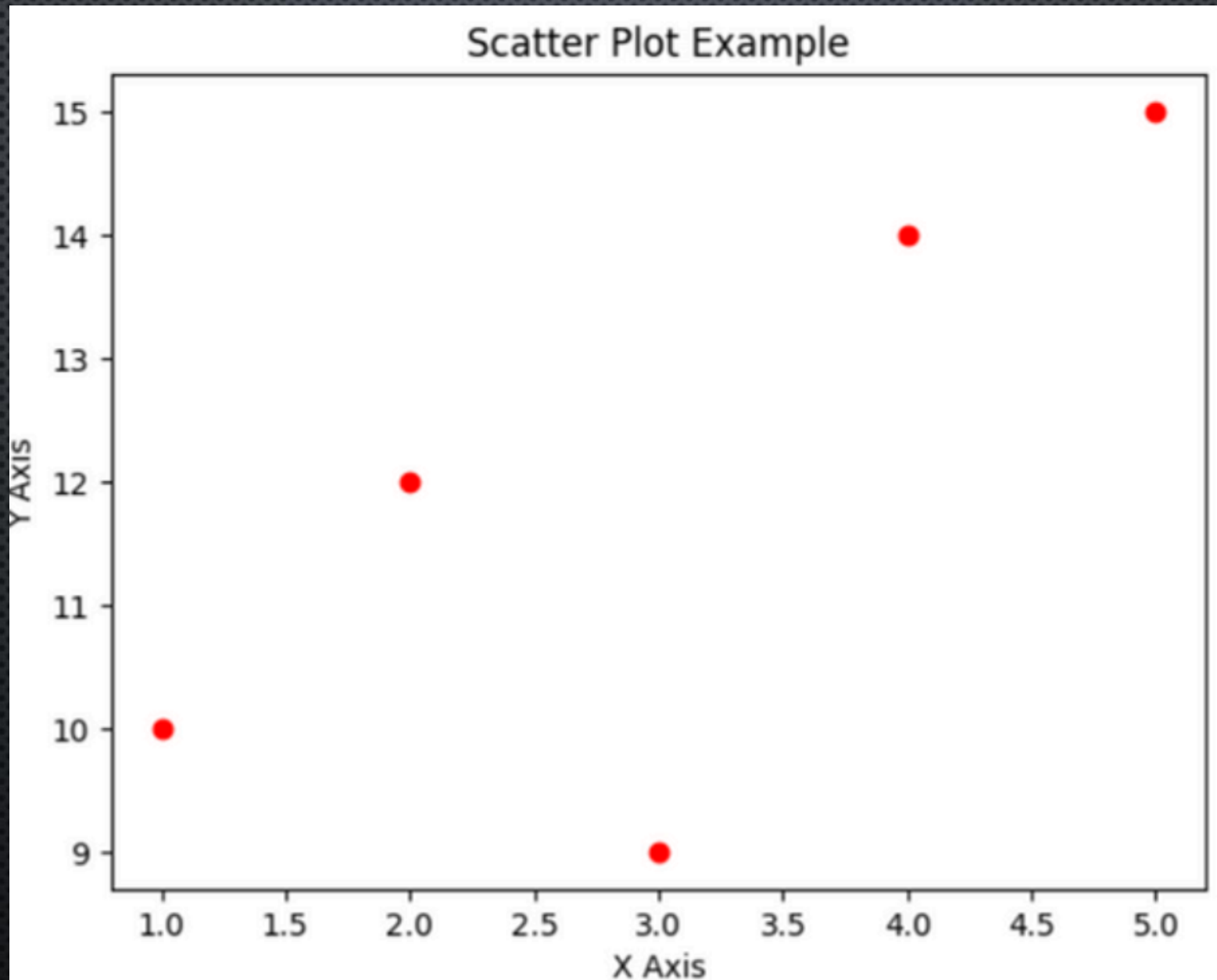


```
import matplotlib.pyplot as plt # Import the Matplotlib library

# Data
x = [1, 2, 3, 4, 5] # X-axis data
y = [10, 12, 9, 14, 15] # Y-axis data

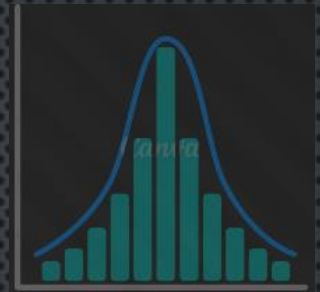
# Create Scatter Plot
plt.scatter(x, y, color='red') # Create a scatter plot with red dots
plt.title('Scatter Plot Example') # Add a title to the plot
plt.xlabel('X Axis') # Label for the X-axis
plt.ylabel('Y Axis') # Label for the Y-axis
plt.show() # Display the plot
```

SCATTER PLOTS



Use Case: Showing the distribution of data.

HISTOGRAMS



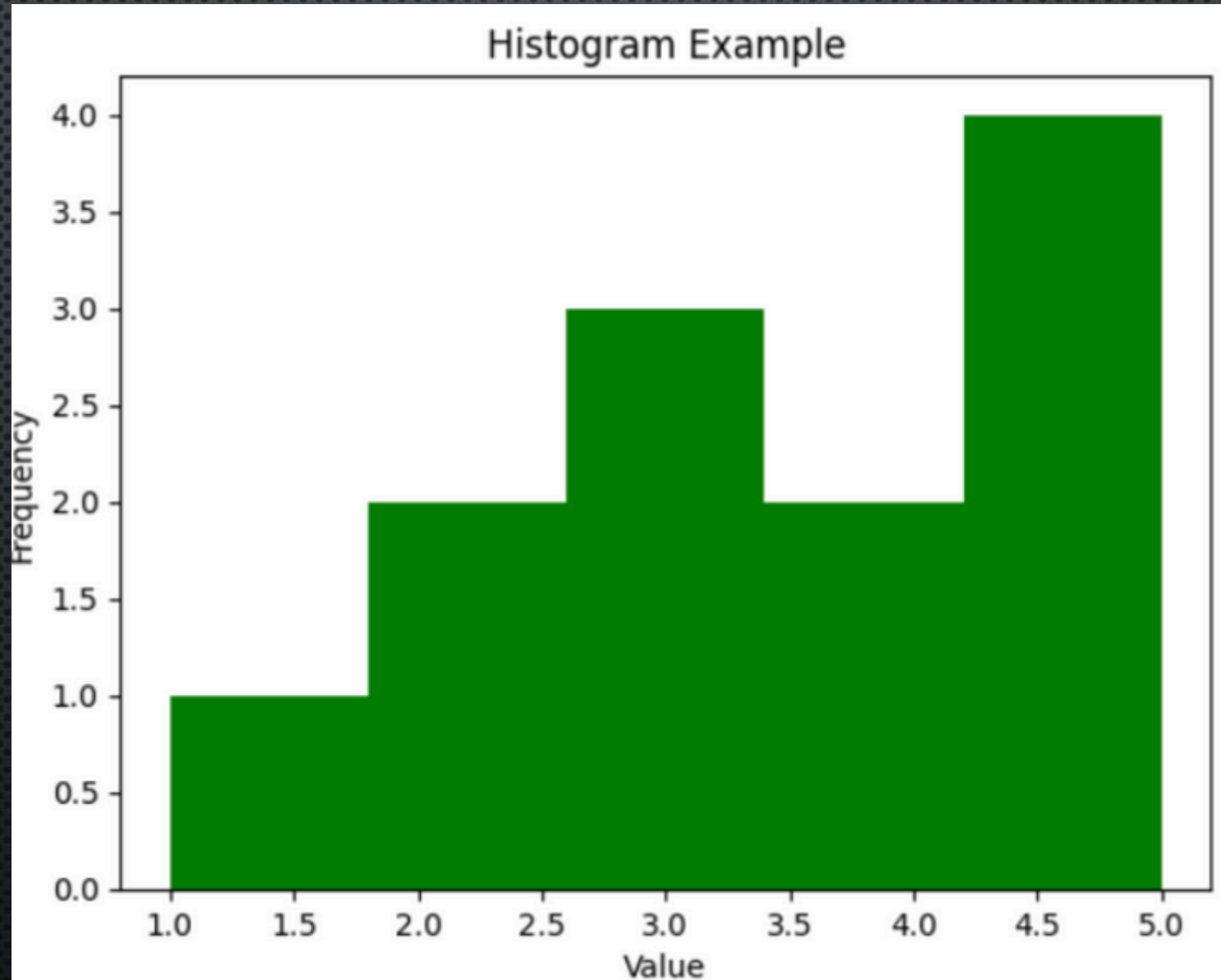
Library: Matplotlib

```
import matplotlib.pyplot as plt # Import the Matplotlib library

# Data
data = [1, 2, 2, 3, 3, 3, 4, 4, 5, 5, 5, 5] # Sample data to plot

# Create Histogram
plt.hist(data, bins=5, color='green') # Create a histogram with 5 bins and green bars
plt.title('Histogram Example') # Add a title to the plot
plt.xlabel('Value') # Label for the X-axis
plt.ylabel('Frequency') # Label for the Y-axis (frequency of each bin)
plt.show() # Display the plot
```


HISTOGRAM



Use Case: Showing the distribution of data.

Library: Matplotlib

PIE CHART



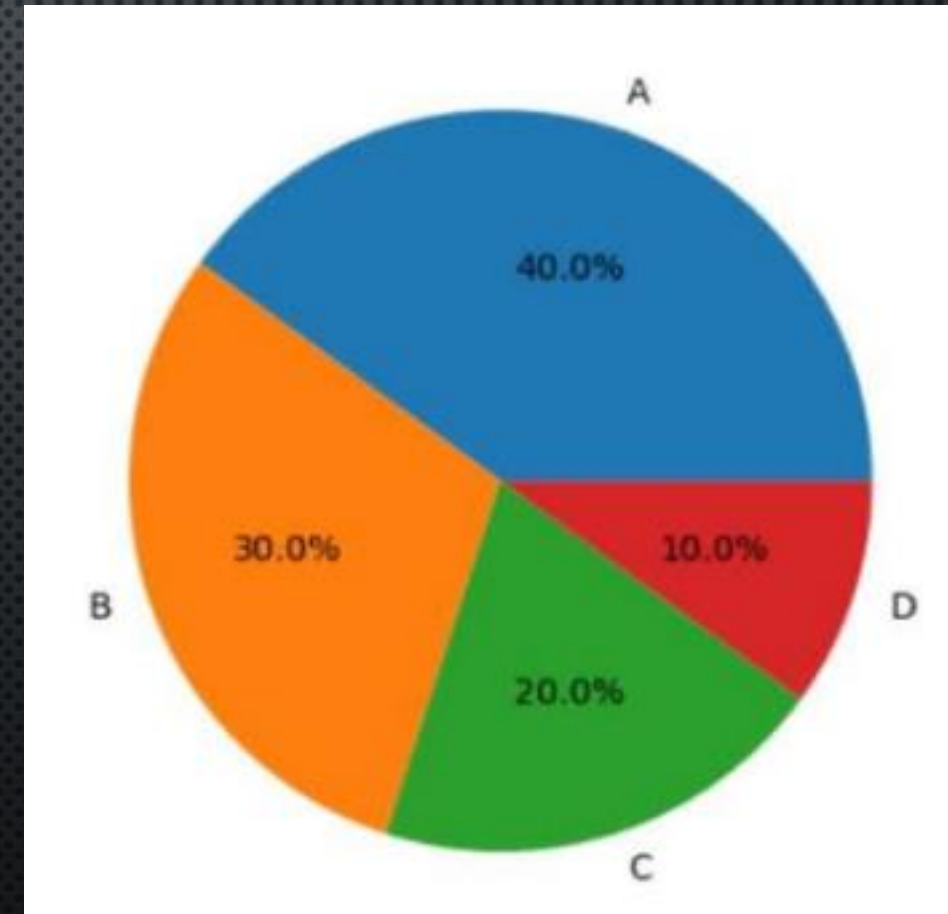
```
import numpy as np
import matplotlib.pyplot as plt

sizes = [40, 30, 20, 10]
labels = ['A', 'B', 'C', 'D']

plt.pie(sizes, labels=labels, autopct='%1.1f%%')

plt.show()
```

https://proclusacademy.com/blog/customize_matplotlib_piechart/



Use Case: Visualizing data in matrix form (correlation, confusion matrix).

Library: Seaborn

HEATMAPS



```
import seaborn as sns # Import the Seaborn library
import matplotlib.pyplot as plt # Import Matplotlib for additional control

# Data
data = [[1, 2, 3], [4, 5, 6], [7, 8, 9]] # Sample 2D array to visualize as a heatmap

# Create Heatmap
sns.heatmap(data, annot=True, cmap='coolwarm') # Create heatmap with annotations and color map
plt.title('Heatmap Example') # Add a title to the heatmap
plt.show() # Display the heatmap
```


Use Case: Showing the distribution of data.

Library: Matplotlib

DONUT CHART



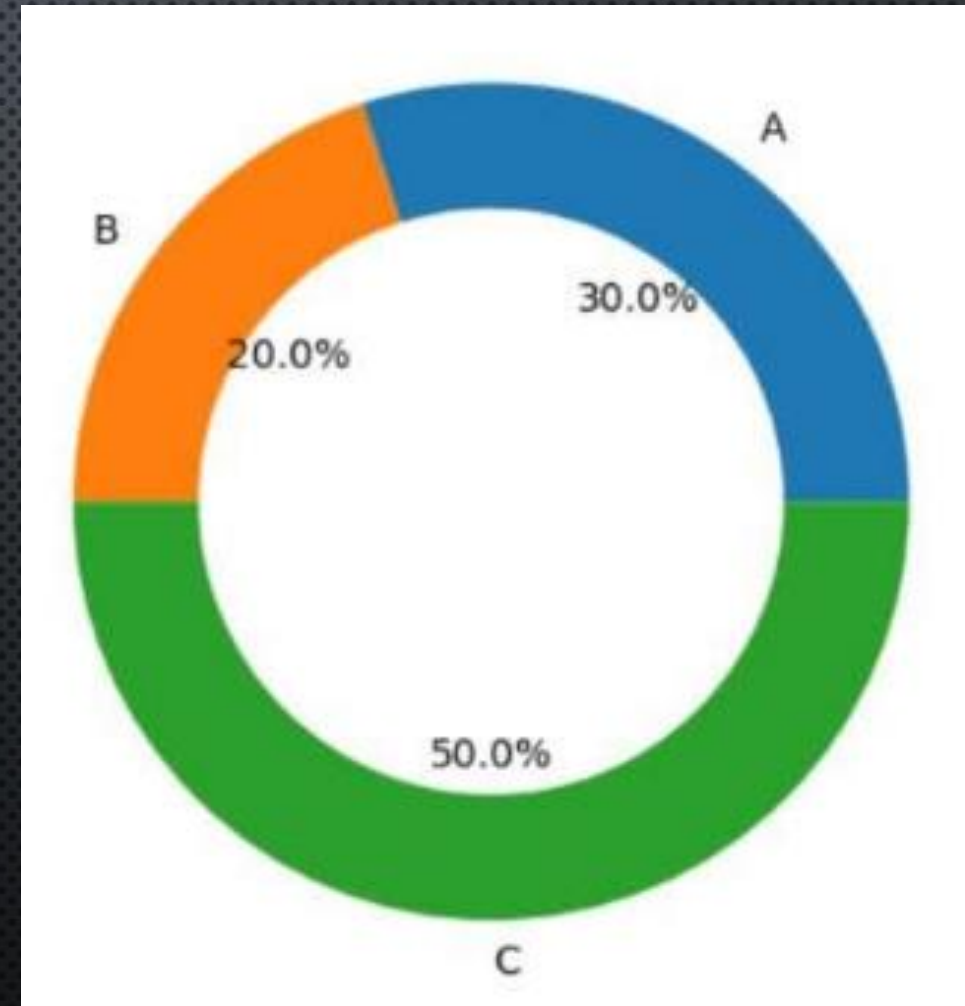
```
import matplotlib.pyplot as plt

sizes = [30, 20, 50]
labels = ['A', 'B', 'C']

plt.pie(sizes, labels=labels, autopct='%1.1f%%',
        wedgeprops={'width': 0.3})
plt.show()
```

<https://python-graph-gallery.com/donut-plot/>

https://matplotlib.org/stable/gallery/pie_and_polar_charts/pie_and_donut_labels.html



Use Case: Visualizing data in matrix form (correlation, confusion matrix).

Library: Seaborn

HEATMAPS

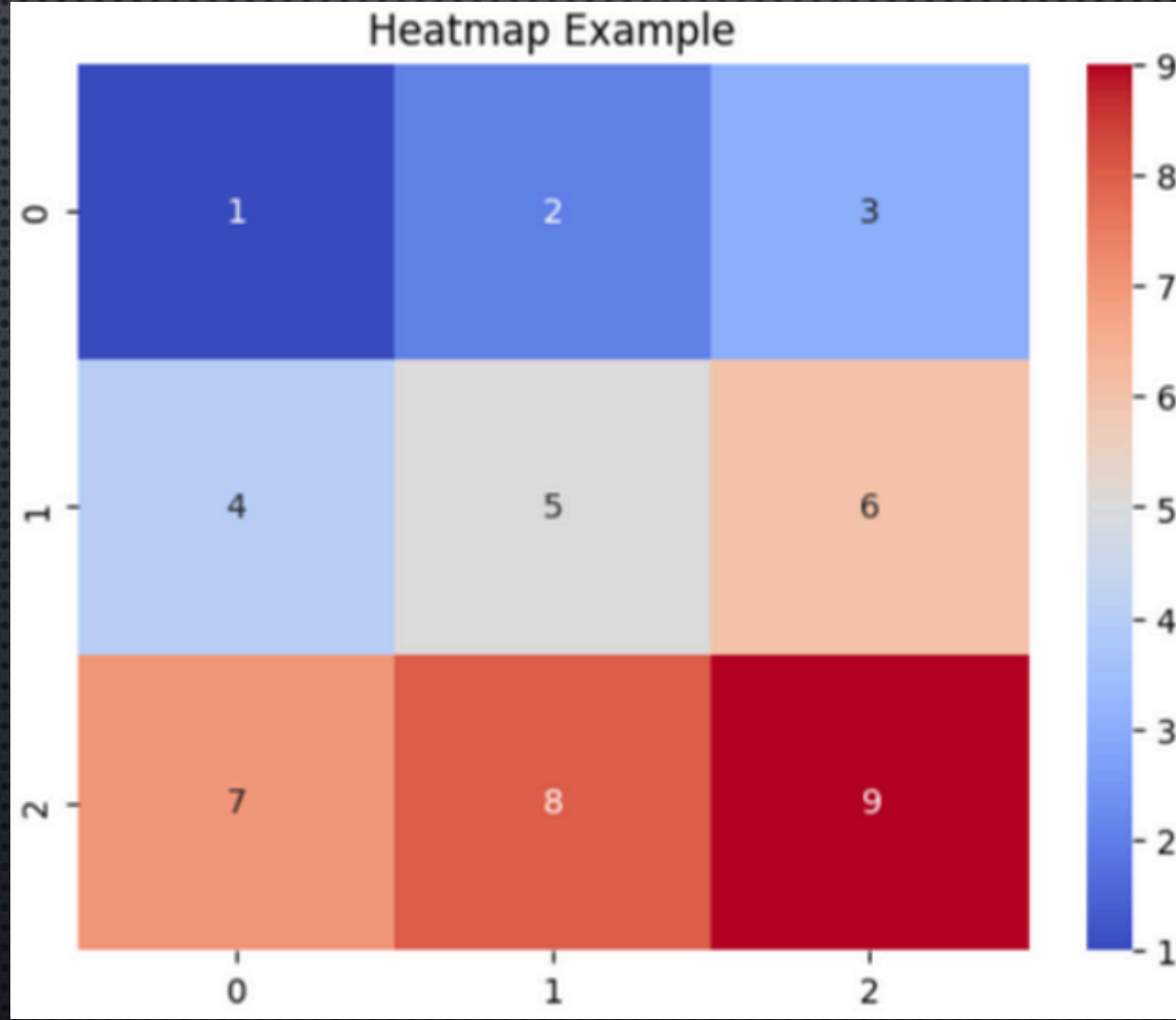


```
import seaborn as sns # Import the Seaborn library
import matplotlib.pyplot as plt # Import Matplotlib for additional control

# Data
data = [[1, 2, 3], [4, 5, 6], [7, 8, 9]] # Sample 2D array to visualize as a heatmap

# Create Heatmap
sns.heatmap(data, annot=True, cmap='coolwarm') # Create heatmap with annotations and color map
plt.title('Heatmap Example') # Add a title to the heatmap
plt.show() # Display the heatmap
```

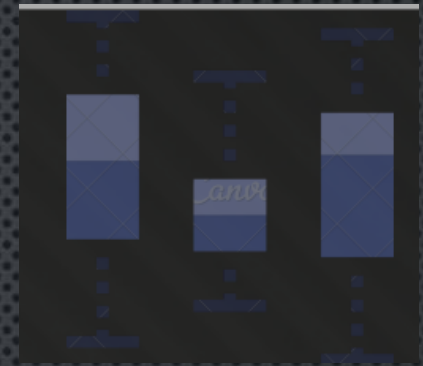

HEATMAP EXAMPLE



Use Case: Summarizing the distribution of a dataset, showing quartiles and outliers.

Library: Seaborn

BOX PLOT

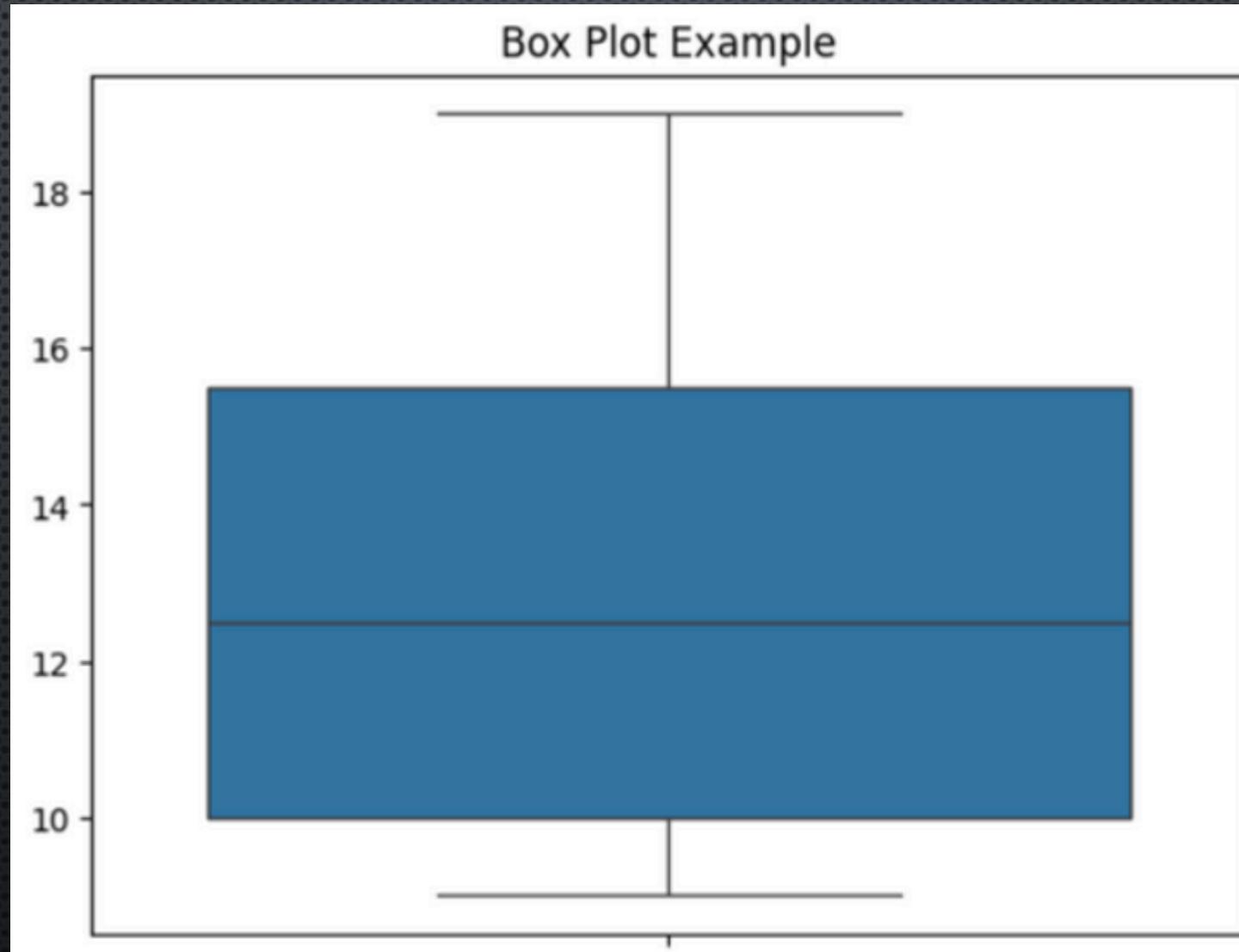


```
import seaborn as sns # Import the Seaborn library
import matplotlib.pyplot as plt # Import Matplotlib for additional control

# Data
data = [10, 12, 15, 13, 17, 19, 10, 9] # Sample data for the box plot

# Create Box Plot
sns.boxplot(data=data) # Create a box plot to show distribution, quartiles, and outliers
plt.title('Box Plot Example') # Add a title to the plot
plt.show() # Display the plot
```

BOX PLOT EXAMPLE



Use Case: Interactive plots for dashboards, complex analysis.

Library: Plotly

INTERACTIVE PLOTS



```
import plotly.express as px # Import Plotly Express for simpler interactive plotting

# Sample Data
df = px.data.iris() # Load built-in Iris dataset

# Create Interactive Scatter Plot
fig = px.scatter(df, x='sepal_width', y='sepal_length', color='species', # Create an interactive scatter plot
                title="Interactive Scatter Plot Example") # Add a title to the plot
fig.show() # Display the interactive plot
```


INTERACTIVE SCATTER PLOT EXAMPLE

